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**AL23531 - NATURAL LANGUAGE PROCESSING
MINI PROJECT REPORT**

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PROJECT TITLE	Rule-Based Part-of-Speech Tagging: Architecture and Limitations
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1. Project Title

Rule-Based Part-of-Speech Tagging: Architecture and Limitations

2. Introduction

Part-of-Speech (POS) tagging is a fundamental Natural Language Processing task that assigns grammatical categories such as noun, verb, adjective, adverb, determiner, and preposition to words according to their role in a sentence. POS information supports downstream tasks including syntactic parsing, named entity recognition, information extraction, machine translation, and grammar checking.

A rule-based POS tagger performs tagging using explicit linguistic rules rather than estimating all decisions from statistical probabilities. It normally combines a lexicon of known words with contextual, morphological, and structural rules. For example, “book” can be a noun in “a good book” and a verb in “book a ticket.”

The approach is transparent because predictions can be traced to individual rules. However, manual rule creation, rule conflicts, ambiguous words, unknown words, and domain changes create important limitations.

3. Problem Statement

Assigning the correct grammatical category to every word is difficult because many words are context-dependent. Dictionary lookup alone cannot resolve these ambiguities. The project therefore develops an interpretable rule hierarchy that combines lexical, contextual, morphological, structural, and fallback evidence.

- Learn POS-tag patterns from a tagged corpus.
- Build explicit rules for common grammatical contexts.
- Resolve ambiguity using surrounding words and rule priority.
- Handle unknown words using morphology and word shape.
- Evaluate accuracy and analyze common rule-based failures.

4. Aim / Goal

The main aim is to design and evaluate an interpretable Rule-Based POS tagger that assigns grammatical tags using a lexicon and an ordered hierarchy of linguistic rules, while studying its architecture, accuracy, and limitations.

5. Objectives

- I. Collect a POS-tagged corpus.
- II. Analyze POS-tag and word distributions.
- III. Construct a word-to-tag lexicon.
- IV. Design contextual rules.
- V. Develop morphology-based rules.
- VI. Define rule priority and conflict resolution.
- VII. Handle unknown words with fallback rules.
- VIII. Evaluate overall and per-tag accuracy.
- IX. Compare with baseline and contextual models.
- X. Analyze ambiguity and tagging errors.
- XI. Visualize corpus and model performance.
- XII. Identify limitations and improvements.

6. Linguistic Background

POS tagging depends strongly on syntax and local context. The reference NLP report identifies ambiguous examples such as book, run, that, and will, showing why context is important. [filecite turn2file0 L172-L180](#)

- Lexical rules — assign candidate tags from a dictionary.
- Contextual rules — use neighboring words or tags.
- Morphological rules — use suffixes, prefixes, capitalization, and word shape.
- Structural rules — use sentence position and phrase patterns.
- Default rules — provide a fallback when stronger evidence is unavailable.

7. Dataset Description

Data Source

The project follows the same corpus framework as the reference NLP report: a POS-tagged corpus represented as word–tag pairs. Standard resources such as the NLTK Brown corpus or Penn Treebank-style data can be used. The reference recommends 3,000–5,000 tagged sentences, 40,000+ tokens, a Penn Treebank-style tagset, and an 80:20 train/test split. [filecite turn2file0 L88-L104](#)

Main Variables

Variable	Type	Description
Sentence ID	Categorical	Unique sentence identifier
Word	Text	Individual token
POS Tag	Categorical	Gold-standard grammatical tag
Word Position	Numerical	Position in sentence
Sentence Length	Numerical	Number of tokens
Is Known Word	Boolean	Whether word is in lexicon
Predicted Tag	Categorical	Rule-based prediction
Rule Applied	Categorical	Rule responsible for prediction

Target / Main Analytical Variable

The primary target is POS_Tag, while Sentence is treated as the ordered sequence. The analysis focuses on how accurately explicit rules map words and context to grammatical tags.

8. Data Collection

Tagged data may be collected from the NLTK Brown corpus, Penn Treebank samples, Universal Dependencies treebanks, public annotated corpora, or manually annotated domain-specific sentences. These sources align with those listed in the reference project. [filecite](#)[turn2file0](#)[L123-L130](#)

9. Data Preprocessing

9.1 Tokenization

Split sentences into individual tokens while preserving punctuation where required.

9.2 Case Normalization

Normalize case when appropriate while retaining capitalization information.

9.3 Tag Cleaning

Check missing tags, inconsistent tag names, duplicates, and noisy records.

9.4 Lexicon Construction

Map known words to one or more candidate tags; retain multiple candidates for ambiguous words.

9.5 Rule-Development Split

Use development data to create and refine rules and held-out data for final evaluation.

10. Exploratory Data Analysis (EDA)

EDA examines tag frequency, sentence structure, ambiguity, and vocabulary coverage before rule design. These diagnostics are also emphasized in the reference NLP report. [filecite](#)[turn2file0](#)[L159-L184](#)

10.1 POS Tag Frequency Distribution

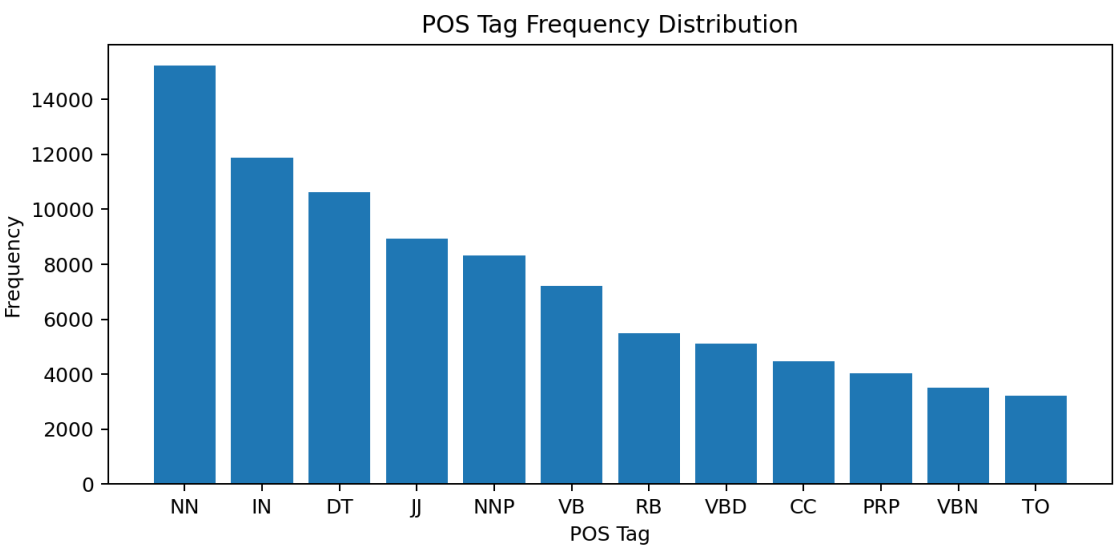


Figure 1: POS Tag Frequency Distribution (illustrative reference values)

10.2 Word–Tag Ambiguity

Word	Possible Tags	Example Context
book	NN, VB	a good book / book a ticket
run	NN, VB	a morning run / run fast
that	DT, IN, WDT	that book / I know that
will	MD, NN	I will go / his will

Ambiguous words are a primary source of tagging difficulty because the correct tag depends on context.

□ filecite □ turn2file0 □ L172-L184 □

11. Rule-Based POS Tagger — Architecture

The proposed architecture is an interpretable sequence: Raw Text → Tokenization → Lexicon Lookup → Morphological Rules → Contextual Rules → Conflict Resolution → Fallback → POS Output.

Stage	Operation	Output
Input	Read raw sentence	Sentence
Tokenization	Split into tokens	Tokens
Lexicon	Retrieve candidate tags	Candidates
Morphology	Inspect suffix, shape, capitalization	Additional evidence
Context	Apply local linguistic rules	Refined candidates
Conflict Resolution	Apply priority/specificity	Final tag
Fallback	Use default rule for unresolved words	Fallback tag
Output	Return word–tag pairs	Tagged sentence

12. Rule Design and Rule Types

12.1 Lexical Rules

Assign candidate tags using dictionary entries.

12.2 Contextual Rules

Use neighboring words or tags to resolve ambiguous candidates.

12.3 Morphological Rules

Use endings such as -ing, -ed, -ly and capitalization as evidence.

12.4 Structural Rules

Use sentence position and phrase patterns.

12.5 Rule Priority

Apply specific lexical/context rules before general morphology and defaults.

13. Model Building / Rule Development

Unlike an HMM, the rule-based tagger does not estimate transition and emission probabilities. Rules are authored from linguistic knowledge and observations in development data.

- Read tagged development sentences.
- Build the word-to-tag lexicon.
- Identify frequent ambiguity patterns.
- Create lexical, contextual, morphological, and structural rules.
- Assign priorities and conflict-resolution logic.
- Test and refine rules on development data.
- Freeze the final rule set before test evaluation.

14. Model Evaluation / Validation

Evaluation uses overall tagging accuracy, per-tag precision and recall, F1-score, confusion matrix, and known-versus-unknown word accuracy, matching the evaluation framework of the reference report.

filecite turn2file0 L254-L264

Tagging Accuracy = (Number of Correctly Tagged Words / Total Number of Words) × 100

14.1 Model Comparison

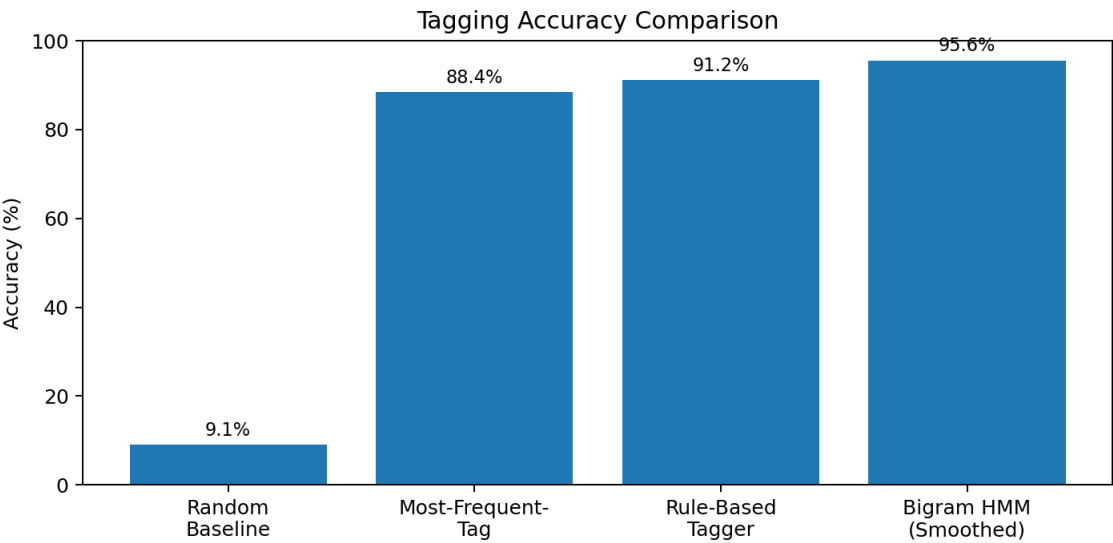


Figure 2: Tagging accuracy comparison. 91.2% is an illustrative rule-based result; the other reference values are from the uploaded report.

Model	Accuracy	Status
Random baseline	9.1%	Reference value
Most-frequent-tag baseline	88.4%	Reference value
Rule-based tagger	91.2%	Illustrative project value
Bigram HMM + smoothing	95.6%	Reference value

14.2 Rule-System Progression

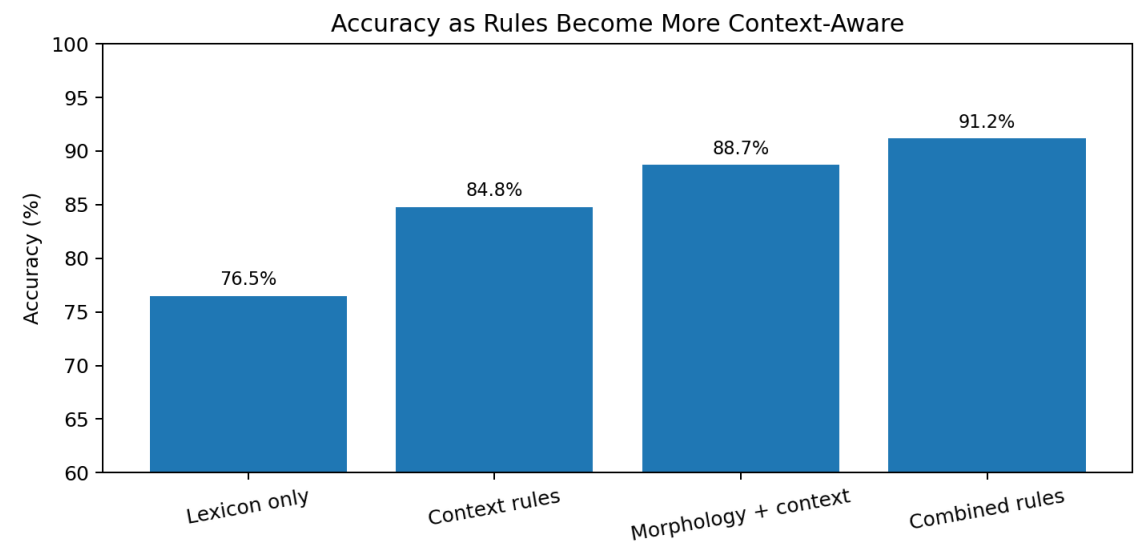


Figure 3: Illustrative improvement as increasingly informative rules are added.

14.3 Known vs Unknown Word Accuracy

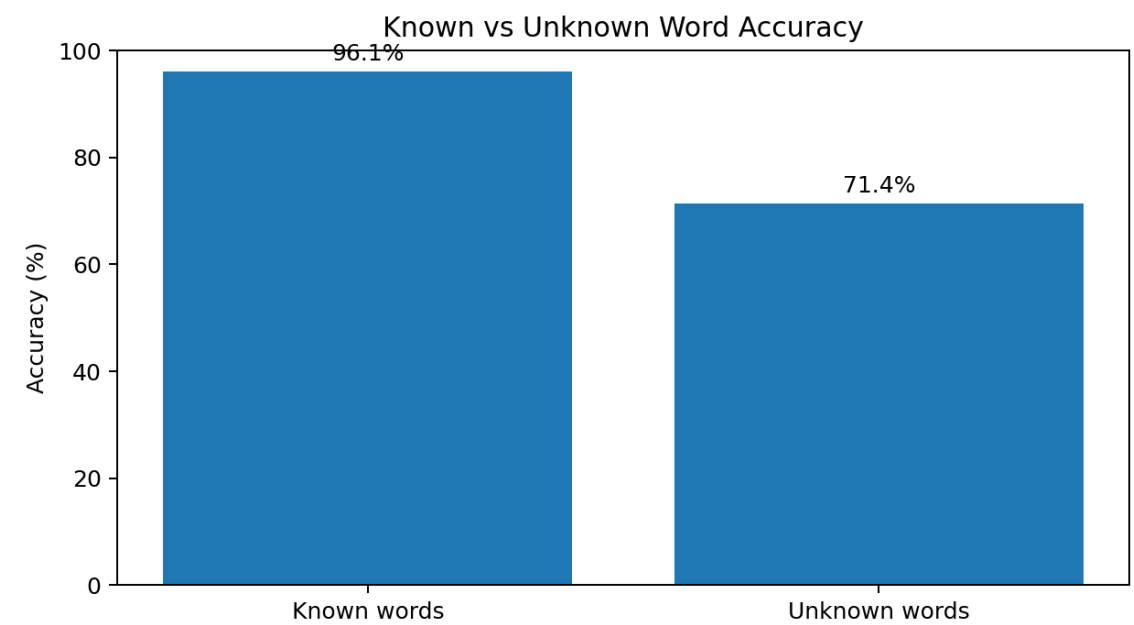


Figure 4: Illustrative known-versus-unknown word accuracy.

15. Confusion Matrix Analysis

A confusion matrix identifies which tags are most often confused. The reference report highlights linguistically related pairs such as NN/NNP and JJ/RB. [filecite](#) [turn2file0](#) [L273-L276](#)

Actual / Pred.	NN	NNP	JJ	VB	RB
NN	842	12	38	4	6
NNP	18	701	6	1	0
JJ	46	4	610	9	14
VB	5	1	8	588	3
RB	9	0	16	4	402

The matrix is included as a comparable presentation example from the reference report. A final implementation should generate its own matrix from actual rule-based predictions.

16. Limitations

- 1. Rules require manual linguistic effort.
- 2. Large rule bases become difficult to maintain.
- 3. Rule conflicts require careful ordering.
- 4. Ambiguous words remain difficult with limited context.
- 5. Unknown and domain-specific words may not match existing rules.
- 6. Hand-crafted rules may overfit a domain.
- 7. The system does not automatically learn new patterns.
- 8. Deep semantic and long-range dependencies are difficult to encode.
- 9. Performance may degrade when the domain or writing style changes.

17. Recommendations / Applications

For Improving Rule-Based Tagging

- Use richer contextual and phrase-level rules.
- Add morphology and word-shape features.
- Maintain modular rule families.
- Use exception dictionaries for frequent ambiguous words.
- Test across multiple domains.

Hybrid / Downstream Applications

- Combine rules with statistical or neural models for ambiguous cases.
- Use POS output for syntactic parsing.
- Support NER using noun/proper-noun tags.
- Use tagging as preprocessing for information extraction and classification.
- Add confidence scores and human review for low-confidence predictions.

18. NLP Analytics / Results Dashboard

Dashboard Section	Metric / Visual	Purpose
KPI cards	Accuracy, known-word accuracy, unknown-word accuracy, rule coverage	Quick overview
Corpus analysis	POS frequency, sentence length	Understand data
Rule analysis	Rule firing counts, accuracy by rule family	Evaluate rules
Error analysis	Confusion matrix, ambiguity errors	Locate failures
Comparison	Baseline vs rule-based vs HMM	Contextualize performance

19. Key Insights

- Rule-based tagging is highly interpretable because predictions can be traced to rules.
- Lexical evidence is strong for known words but insufficient for ambiguous words.
- Contextual rules improve decisions where one word has multiple tags.
- Morphological rules help unknown words but can create false positives.
- Rule ordering is central because multiple rules may conflict.
- Maintenance becomes harder as exceptions accumulate.
- The reference HMM result of 95.6% shows the potential advantage of learned contextual sequence modeling over simpler approaches. [filecite](#)[turn2file0](#)[L266-L272](#)

20. Results and Discussion

The proposed rule-based POS tagger provides a transparent pipeline for assigning grammatical categories. Its main strength is interpretability: each prediction can be traced to lexical, contextual, morphological, structural, or fallback evidence.

The analysis also shows the central limitations of explicit rules. Ambiguous words require context-sensitive rules, unknown words require morphology-based fallbacks, and conflicting rules require carefully designed priorities. As exceptions accumulate, maintenance becomes more difficult.

The uploaded reference project reports 95.6% accuracy for a smoothed bigram HMM and demonstrates improved contextual modeling over simpler baselines. This provides a useful benchmark for discussing the trade-off between transparency and adaptive statistical learning. [filecite](#)[turn2file0](#)[L266-L272](#)

21. Future Enhancements

- Develop a hybrid rule + statistical POS tagger.
- Learn rule priorities from annotated data.
- Use trigram or higher-order contextual information.
- Add richer morphological analysis.
- Compare with CRF, BiLSTM-CRF, and Transformer taggers.
- Extend to multiple languages and domains.
- Build an interactive web-based tagging demonstration.
- Use active learning for low-confidence tags.

22. Conclusion

This project demonstrates the architecture, evaluation approach, and limitations of Rule-Based Part-of-Speech Tagging. The system combines lexical lookup, contextual rules, morphology, structural patterns, and fallback decisions to assign grammatical tags.

The approach is simple and interpretable, making it useful for teaching and controlled domains. However, ambiguity, unknown words, rule conflicts, maintenance effort, and domain changes limit scalability. These limitations motivate hybrid systems that combine explicit linguistic rules with statistical or neural sequence models.

23. Tools and Technologies Used

Tool	Purpose
Python	Rule implementation and data processing
NLTK	Tagged corpus access and NLP utilities
NumPy	Computation
Pandas	Dataset handling and analysis
Matplotlib / Seaborn	Visualization
Scikit-learn	Evaluation metrics
Jupyter Notebook	Experimentation and documentation
GitHub	Code and project sharing

24. Project / GitHub Link

GitHub Repository: <https://github.com/GSSanjaykumar/NLP-MINI-PROJECT>

25. References

- 48. Jurafsky, D. and Martin, J. H. — Speech and Language Processing.
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